Assignment-2

Group-036

1 Task 1 - Slate Health Check

This section provides a high-level overview of the financial health and composition of the studio's historical slate.

Some cleaning was required to address missing or implausibly low budget and revenue values. While most dollar amounts were recorded in full, some appear to have been entered in hundreds of thousands or millions. For example, *Escape from Tomorrow* has a recorded budget of 650, compared with its actual reported budget of $650,000.

Given the initial dataset size ((n = 4803)), we excluded films with budgets or revenues below 1,000 and budgets above USD 550 million, the largest reported film budget. We also excluded films released before 1974, as franchise labels were never applied to films before then, making comparisons between Original and Franchise films impossible.

This left 3060 films for analysis. Franchise films accounted for 7.2% of the cleaned dataset, compared with 5.4% of the original. We consider this difference small enough that it is unlikely to meaningfully skew our analysis.

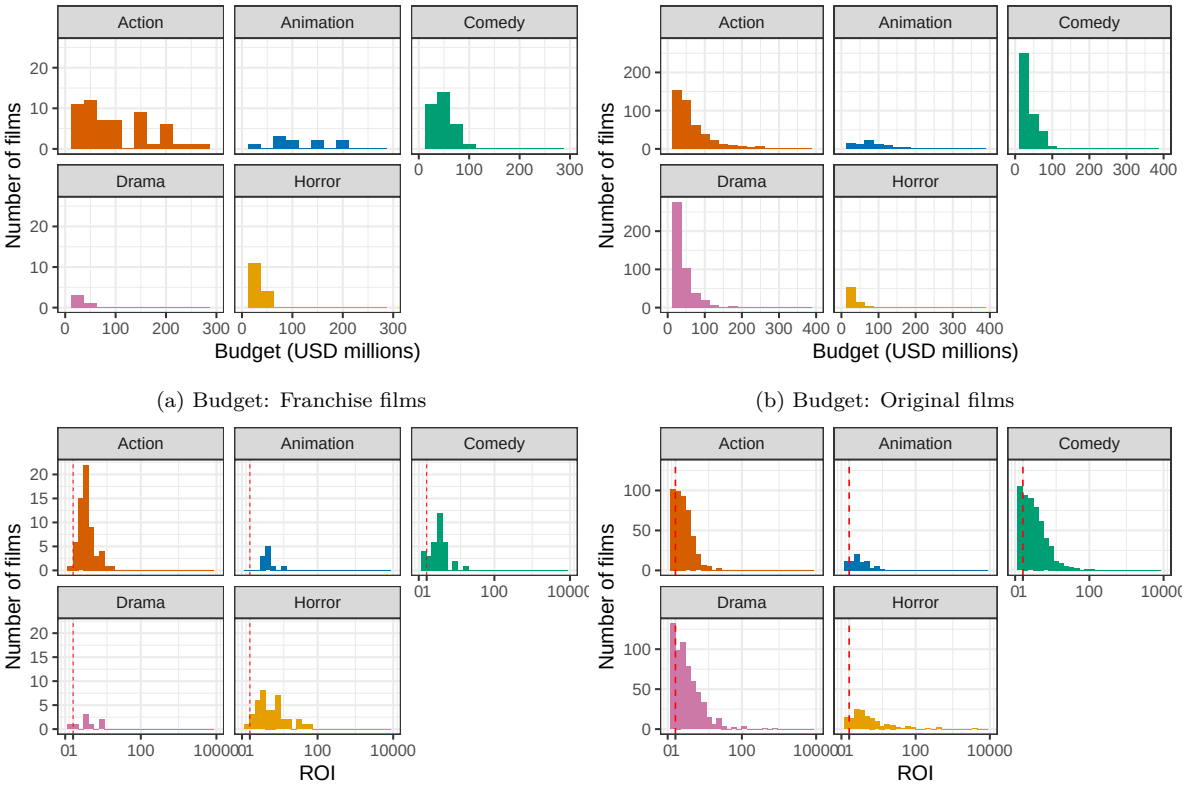
1.1 Summary of data

> **Note 1: Coding for Original vs Franchise**
>
> First films in a franchise are labelled as "Original" rather than as part of a franchise, and are therefore included in the data for the "Original" variable (see Table 1, where *Paranormal Activity* is an Original film, and *Paranormal Activity 2* is a Franchise film.)
> Additionally, some films that are clearly part of a franchise are not labelled as such; i.e. all 4 Pirates of the Caribbean movies are labelled as Original movies.
> As such, the figures in Table 1 should be taken as *indicative* of overall trends only, not treated as absolute truths.

Table 1: Summary statistics for the historical slate

Statistic	Franchise	Original
Number of movies	221	2839
Most Common Genre	Action (n = 63)	Drama (n = 681)
Mean Production Budget (USD millions)	65.15	40.88
Median Production Budget (USD millions)	40	27
Production Budget Standard Deviation (USD millions)	63.2	42.48
Mean Return on Investment (ROI)	4.84	11.05
Median ROI	3.36	2.13
ROI Standard Deviation	6.11	254.93
Max Production Budget (USD millions)	280 *(Avengers: Age of Ultron)*	380 *(Pirates of the Caribbean: On Stranger Tides)*
Minimum Production Budget (USD)	852,510 *(4 Months, 3 Weeks and 2 Days)*	7,000 *(Primer)*
Best ROI	59.17 *(Paranormal Activity 2)*	12890.39 *(Paranormal Activity)*
Worst ROI	0 *(5 Days of War)*	0 *(The Adventurer: The Curse of the Midas Box)*

1.2 Budget and ROI plots



(a) Budget: Franchise films

(b) Budget: Original films

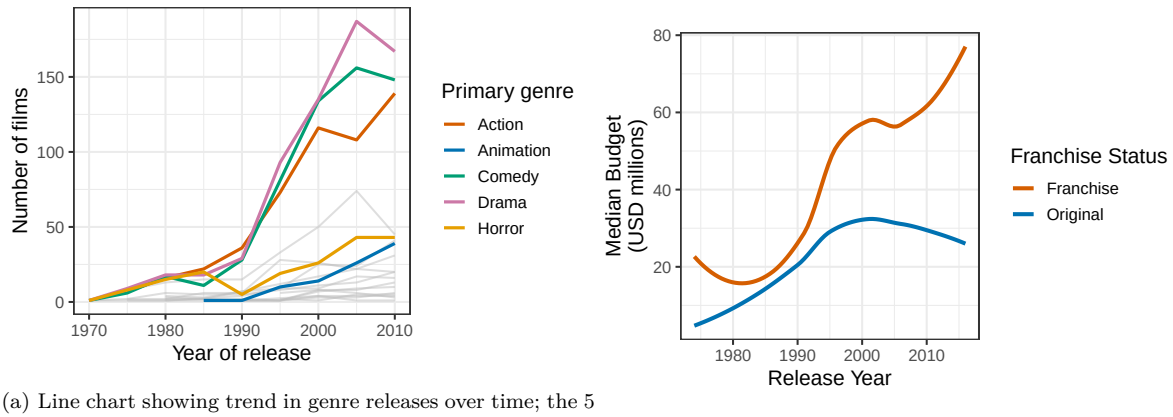
(c) ROI: Franchise films

(d) ROI: Original films

Figure 1: Histograms showing distribution of budget costs and ROI for Original and Franchise films, faceted by film genre; only Meridian's 5 focused genres are included

1.3 Slate composition over time

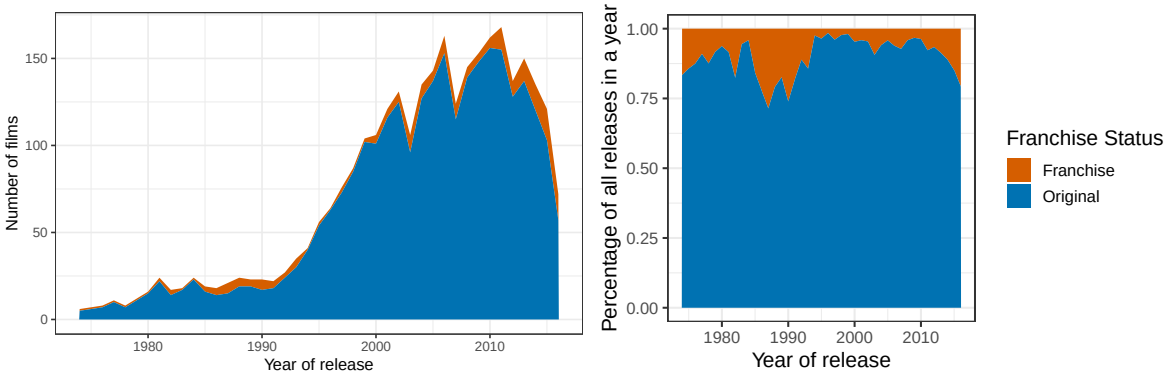
Although we have some data from 2015 on, it is not complete data and is therefore not included in Figure 2 as it skews visual temporal analysis. It is unclear if the data we have from 2014 is fully complete, but it has been included until we are given a compelling reason to exclude it.



(a) Line chart showing trend in genre releases over time; the 5 genres Meridian is focused on are highlighted; film release dates have been aggregated into 5 year blocks to make trends more visible

(b) Smoothed line plot showing median budget for Original vs Franchise films over the last ~50 years

Figure 2: Line charts showing changing composition of slate over time



(a) Stacked area chart displaying actual numbers of films for each year

(b) Filled area chart showing franchise status as a percentage of all films released in a given year

Figure 3: Area charts showing the split between original and franchise films by year

1.4 Summary

Sequels are generally more expensive to make than original films, and are generally a safer bet than original films; they have a higher ROI on average, and are less likely to fall below an ROI of 1 (meaning they generally at least make their money back) (Table 1 and Figure 1).

However, original films have much higher variability in ROI and may represent a riskier but more lucrative investment strategy.

The three genres that have consistently seen the most films released over the last 25+ years are Action, Comedy and Drama films (Figure 2a), all of which Meridian is invested in.

There was a significant spike in the relative amount of Franchise films being released around 1990, which then completely flattened out prior to 2000 (Figure 3b). With the recent trend towards a higher percentage of releases being Franchise movies, we should look towards that period in the 90s when developing our strategy for the next decade. This will be discussed more in Section 2.

2 Task 2

Figure 1 suggests that the budget and ROI data we have is heavily right-skewed. We should first make sure that Simpson's paradox isn't at play, and that these distributions are accurate when the films in the dataset are grouped by franchise status.

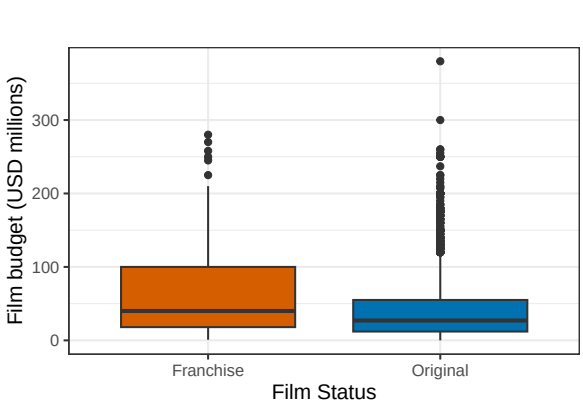
2.1 Full Budget and ROI distributions of Franchise and Original films

Figure 4 shows that yes, this right-skewed distribution does hold when we group films by franchise status. This will make the median more useful than the mean when comparing budgets, as it more accurately represents the size of budget we might expect a film to have based on genre or franchise status.

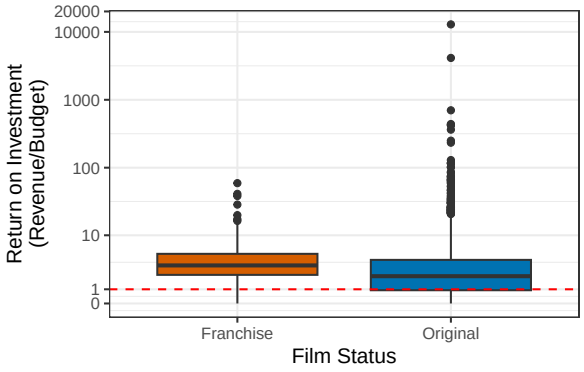
The distributions and comparisons between groups follow what we might expect from our previous look at the summary statistics of the dataset (Table 1). Franchise films have a higher median budget and ROI but Original films exhibit higher kurtosis in both plots, with more pronounced tails and a greater likelihood of extreme values. Both franchise status categories are bounded at 0 in both statistics as we would expect, as we can't have a negative budget or ROI. The much tighter clustering of Franchise film ROIs above 1 indicates that any money invested by the studio into a higher budget Franchise film will likely be earned back, while the lower quartile for Original film ROIs sits at 1, meaning at least a quarter of all Original films in the dataset did not make back their budget.

2.2 Calculating confidence interval for differences in median budgets

As shown in Figure 4 and Table 1, Original films have a lower median budget than Franchise films. To assess the uncertainty in this difference, we use bootstrapping to estimate a 95% confidence interval for the difference in median budgets.



(a) Budget



(b) ROI; dashed red line is at ROI = 1 i.e. films above the line made a profit, films below lost money

Figure 4: Boxplots showing distribution of budgets and Return on Investment (Revenue divided by Budget) for Original vs Franchise films

Each bootstrap samples (n) observations with replacement, where (n = 3060), the size of the original dataset. As Minh noted in Seminar 6, *"probably the word with replacement is probably the most important part with bootstrapping"*. Replacing each sampled observation allows it to be selected multiple times, introducing variation between samples and approximating sampling uncertainty.

We generate 5,000 bootstrap samples and calculate the difference in median budgets for each. The 2.5th and 97.5th percentiles of these estimates form the 95% confidence interval. The observed difference in median budgets is 13 million USD.

The bootstrapped 95% confidence interval ranges from 8 to 23 million USD. As the interval lies entirely above zero, the results suggest that Franchise films have a higher median budget than Original films at the conventional 5% significance level. The point estimate of 13 million USD falls within the interval, making it consistent with the bootstrapped distribution. Figure 5 illustrates this distribution.

2.3 Have we verified leadership's concern?

In one word: yes.

In more words: kind of. While there is a real difference in median budget between Franchise and Original films with the Franchise slate tending to absorb larger budgets, it would not be accurate to say that the Original slate is mostly full of low-budget projects. The median cost for a franchise film in the dataset is $40m, and based on our bootstrapped 95% confidence interval the median Original film has a budget between $8m - $23m lower than that. That means the median Original film is somewhere between ~25% and ~50% cheaper than the median Franchise

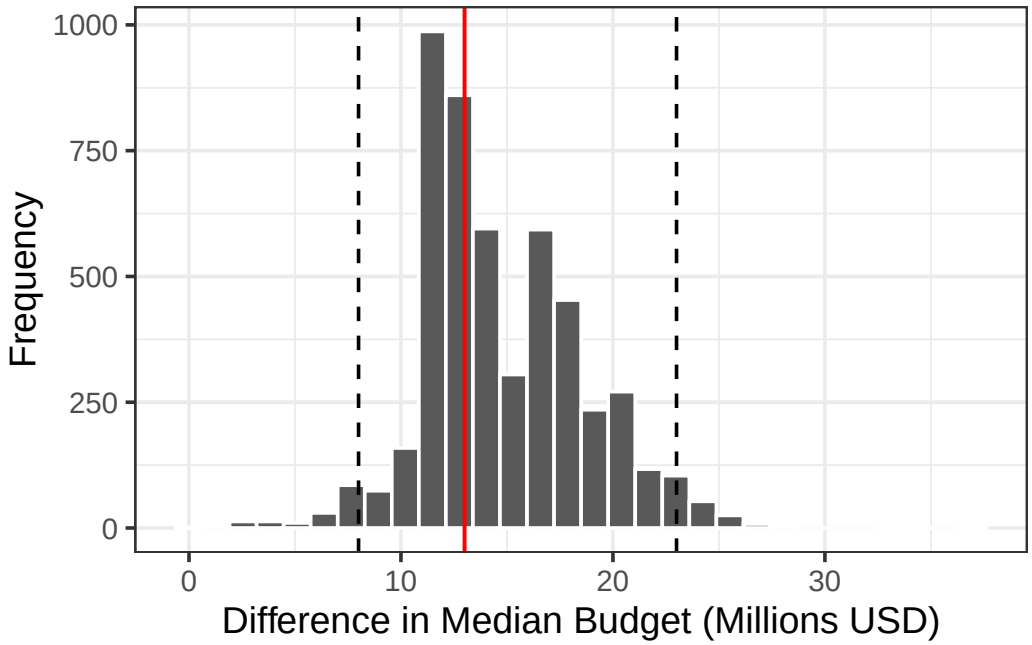


Figure 5: Bootstrap Distribution: Difference in Median Budgets, Franchise Median - Original Median (Solid line = observed, Dashed = 95% CI)

film, and as seen in Figure 4a, the most expensive Original films actually cost more than the most expensive Franchise films. Given the higher median cost of Franchise films and the stable ROI for the Franchise slate seen in Figure 4b, leadership would be correct to assume that the Franchise slate is the *stable* part of studio revenue. But the variability in ROI for the Original slate combined with the fact that in absolute dollar value the median Original film does not represent a significantly smaller investment for a studio suggests that a hit Original film could easily generate a large percentage of studio revenue for an annual period.

2.4 Advice

Meridian should consider the cyclical pattern of Franchise releases when planning its investment strategy for the next decade. As shown in Figure 3b, Franchise films peaked as a share of the total slate in the late 1980s and early 1990s, before falling to their lowest recorded share by the late 1990s. Their share has since risen to nearly match the earlier peak. If this pattern repeats, the next decade could see another decline as audiences experience Franchise fatigue and demand Original films.

Additionally, Figure 4b and Figure 1d highlight the ROI potential of Original films across Meridian's focus genres. Horror is particularly promising, combining the second-lowest flop

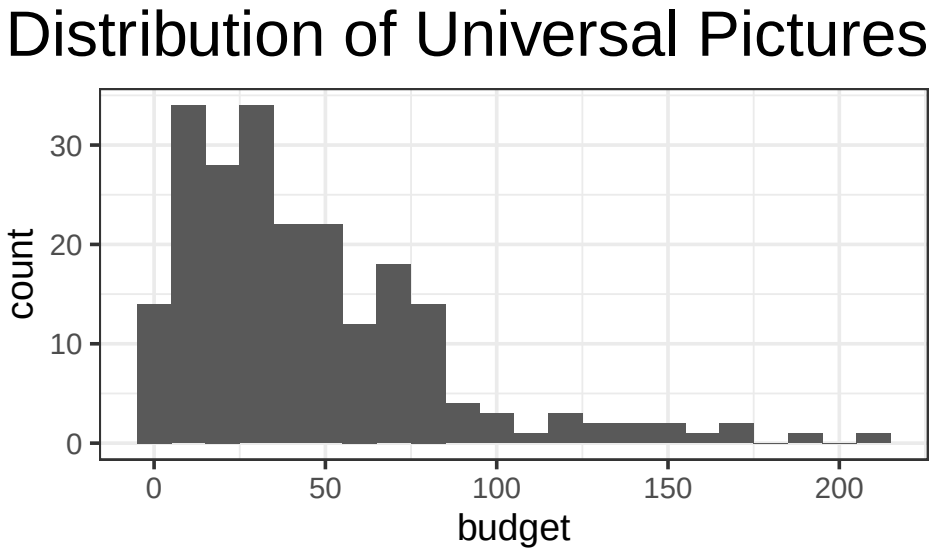
rate with substantial potential returns. Original horror films may therefore offer an attractive balance between investment risk and profitability.

As such, we recommend that Meridian strategically prioritise Original horror films in its 2017 release slate, while gradually experimenting with Original films in other high-potential genres. By carefully monitoring performance and adjusting budget allocations in response to observed returns, Meridian can progressively refine its investment strategy, mitigate the risks associated with Franchise fatigue, and improve its potential for sustained profitability over the next decade.

3 Task 3

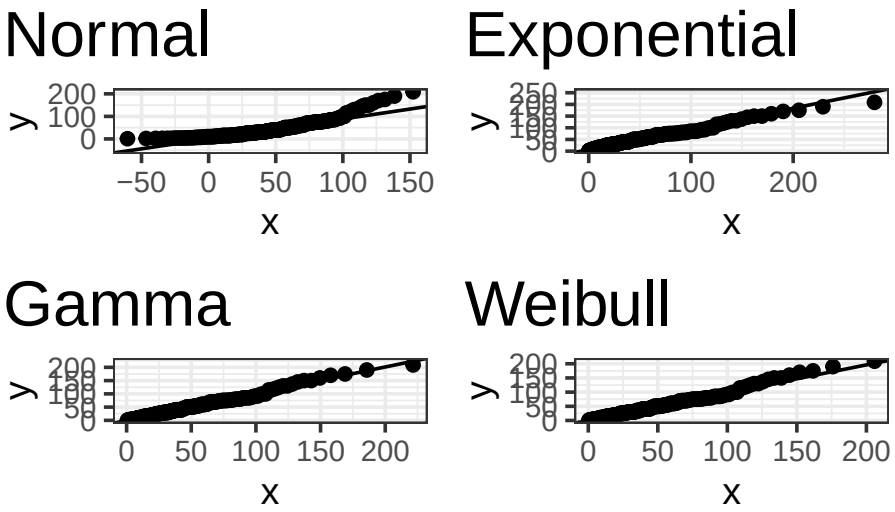
3.1 Part 1: Production Budget Modelling

With the goal to construct a model that predicts an appropriate movie budget distribution, a competitor budget analysis was conducted as a credible reference point in line with industry standards. This was performed by sampling the budgeting model from Universal Pictures Studios, which is a direct industry competitor, and fitting a realistic probability model for Meridian to follow after similar data processing from previous analysis.



After data processing, the budget distribution is right skewed and only contain positive values. Therefore, Exponential, Gamma and Weibull distribution were selected for model evaluation.

model	df	AIC
budget_gamma	2	2108
budget_weibull	2	2109
budget_exp	1	2126
budget_normal	2	2223



The QQ-plot indicates normal was the worst fit as the tails deviated the most from the reference line. The other models had a similar fit as the right tail have a similar distance from the reference line and most of the points overlap with the line.

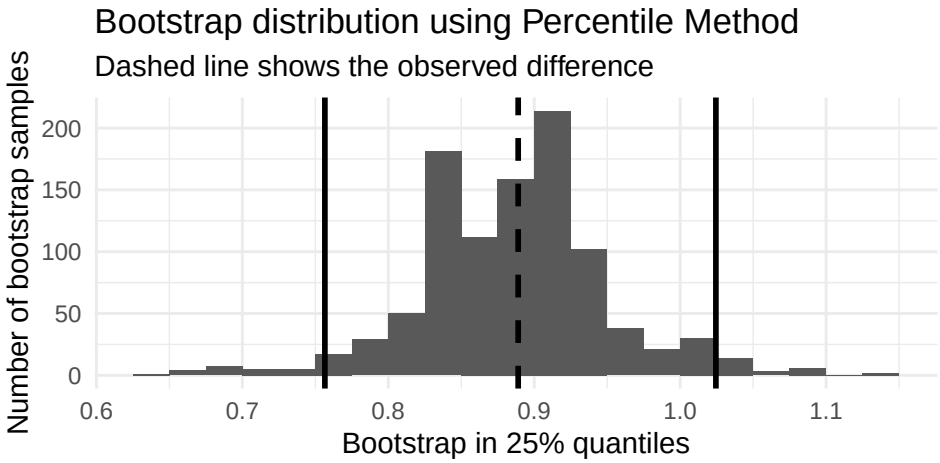
The Gamma model has lowest AIC, which paired with the fitting of the QQ-plot, suggests that it is the best at fitting budget data. However, the Weibull model would also be sufficient as the AIC is slightly larger than the gamma distribution by 1.

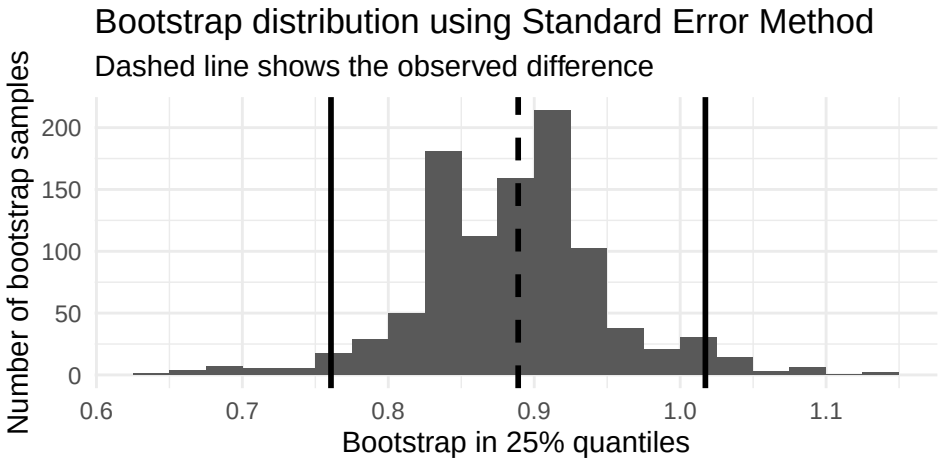
3.2 Part 2: Downside Risk Analysis for Action Movie ROI

Action movies have been established as capital-intensive to produce, which can be proved as risky for budget investment in order to yield significant revenue. As a result, risk analysis into the bottom performing 25% movie ROI was performed across the entire film industry to model conservative target ROI to aim for,

The observed 25% percentile of Action Movie ROI was 0.89

Calculation_method	Lower_Bound	Upper_bound
Percentile bootstrap	0.76	1
Bootstrap SE (t)	0.76	1





Both methods to calculate the confidence interval use the bootstrapped sample, which assumes the sample of Action films is representative of the population. Overall, both the percentile interval and the standard error interval equal to (0.76, 1.00).

The percentile method does not consider symmetry as it takes the 2.5th and 97.5th percentiles directly from the bootstrapped distribution, while the bootstrapped standard error method assumes the distribution is symmetric and approximately bell-shaped which places the lower and upper bounds equidistant from the observed 25th percentile (0.89). Both methods

yielded identical intervals due to the bootstrapped distribution appearing to be approximately symmetric around the observed estimate with little skewness. This indicates the symmetry assumption is upheld where the standard error interval relies on this assumption holding as true. Although either method is suitable here, the percentile method is generally preferred because it does not depend on the symmetry assumption.

3.3 Summary

Based on the analysis, it is recommended to select the Gamma distribution for the simulation. It had the best fit for the budget data as it matched the QQ-plot reference line the closest and had the lowest AIC value after evaluating five different models. It captured the right skew in budgets where few movie budgets had more expensive budgets compared to the majority of movies.

In this sample, 25% of action films had an ROI of 0.576 or lower. We are 95% confident the true 25th percentile of the ROI for all action films across all studios lies between 0.463 and 0.679. For a conservative risk figure to plan around, Finance should consider 0.463 (46.3% return) for an underperforming Action film.

4 Task 4

4.1 Defining a flop and justifying logistic regression

The new flop variable only has two possible outcomes, which means it can be modeled by a Bernoulli distribution, where the parameter p measures the probability that a move is a flop. Hence, logistic regression is the appropriate choice as the log function $\log\left(\frac{p_i}{1-p_i}\right)$ ensures the predicted probabilities remain between 0 and 1. The linear regression is not suitable as it assumes outcomes are continuous and normally distributed, while the flop variable only takes 0 and 1. Poisson regression is also not suitable as it is used when the outcome is a count with no upper limit.

4.2 Comparing two methods of logistic regression

4.2.1 Frequentist Regression Equation

$$\begin{aligned}\log\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right) = & -2.241 - 0.678\text{Animation}_i - 0.327\text{Comedy}_i \\ & - 0.001\text{Drama}_i - 1.100\text{Horror}_i + 1.411\text{Original}_i \\ & - 0.017\text{log budget}_i - 0.502\text{cast size}_i\end{aligned}$$

4.2.2 Bayesian Regression Equation

Weakly informative priors were placed on the log-odds scale: Normal(0, 1.5) for the intercept and Normal(0, 1) for each slope.

$$\begin{aligned}\log\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right) = & -2.17 - 0.63\text{Animation}_i - 0.30\text{Comedy}_i \\ & + 0.02\text{Drama}_i - 1.05\text{Horror}_i + 1.32\text{Original}_i \\ & - 0.02\text{log budget}_i - 0.50\text{cast size}_i.\end{aligned}$$

4.2.3 Confidence and Credible Interval Comparison

Using the 5% significant level, franchise status, cast size, and the Animation, Comedy and Horror genres are significantly associated with the odds of a film flopping. The Drama genre and Log budget coefficient are not significantly associated with the odds of a movie flopping due to the p values being greater than the significant level after adjusting for other predictors.

The coefficient odds ratio (OR) interpretation as follows:

- Animation (frequentist OR = 0.51, Bayesian OR = 0.53): Animated films have about half the odds of flopping compared with Action films.
- Comedy (OR = 0.72, 0.74): Comedies have about 26–28% lower odds of flopping than Action films.
- Drama (OR = 1.00, 1.02): There is no meaningful difference in the odds of flopping between dramas and Action films.
- Horror (OR = 0.33, 0.35): Horror films have about 65–67% lower odds of flopping than Action films.
- Original (OR = 4.10, 3.71): Original films have about 3.7–4.1 times the odds of flopping compared with franchise films.
- Log budget (z) (OR = 0.98, 0.98): A one standard deviation increase in log budget is associated with almost no change in the odds of flopping.
- Cast size (z) (OR = 0.61, 0.61): A one standard deviation increase in cast size is associated with a 39% decrease of odds of flopping.

For original films, the 95% confidence interval for the odds ratio is [2.28, 8.16] and the 95% credible interval is [2.18, 6.65]. For cast size, the intervals are [0.52, 0.70] and [0.52, 0.71]. The cast size intervals are almost identical as the large sample size increases the weight of the data compared the weakly informative prior. The Original intervals differ more where the credible interval is narrower and closer to 1 because the Normal(0, 1) prior shrinks the coefficient. It is noteworthy that confidence intervals calculate the true OR in 95% of repeated samples while the credible interval uses the model, priors and data to calculate the posterior probability that the true OR lies within the interval.

Table 2: Confusion matrix showing actual vs predicted flop rate

	Actual	
Predicted	Not flop	Flop
Not flop	756	136
Flop	835	428

Table 3: Sensitivity analysis of model at varying thresholds

Threshold	Accuracy	Sensitivity	Specificity	Precision	F1
0.50	0.74	0.000	1.00	NaN	NaN
0.40	0.74	0.062	0.98	0.59	0.11
0.30	0.63	0.612	0.64	0.38	0.47
0.26	0.55	0.759	0.48	0.34	0.47
0.10	0.33	0.972	0.10	0.28	0.43

4.2.4 Tested Posterior probability statement: Calculate the posterior probability that the odds ratio for log budget is greater than 1.2 (20% increase in the odds of a flop) or less than 0.8 (20% decrease).

Given the data and our prior, the probability that a 1 standard deviation increase in log budget increases the odds of a flop by more than 20% is 0.0005, and the probability that it decreases the odds by more than 20% is 0.

4.2.5 Confusion Matrix Analysis

See Table 2.

4.2.6 Sensitivity Analysis of Risk Threshold

As only 26% of films were flops, a 0.5 threshold predicts no flops at all (sensitivity of 0), so the model is useless at this threshold because flops are not flagged. A threshold of 0.4 identifies only 6% of flops, which is too low given that approving a film that flops costs far more than rejecting a film that would have succeeded financially. Lowering the threshold to 0.1 catches 97% of flops but flags almost every film, however the specificity of 0.10 and accuracy of 0.33 would not approve enough safe films. At a threshold of 0.26, the model correctly identifies 76% of flops (sensitivity 0.76), but only 48% of non-flops are correctly cleared (specificity 0.48), and only 34% of flagged films actually flop (precision 0.34). Overall accuracy falls from 0.74 to 0.55, which is expected when prioritising the rarer outcome. A threshold of 0.3 gives a more

Table 4: Using our model to asses the flop probability of 3 hypothetical pitches

Pitch	Genre	Type	Budget (USD millions)	Cast size	Flop risk	Lower 95%	Upper 95%	Decision
Pitch A	Action	Franchise	20.0	30	9.7%	5.4%	15.1%	Not flagged
Pitch B	Action	Original	20.0	30	27.9%	24.0%	32.0%	Flop risk
Pitch C	Comedy	Original	20.0	30	22.2%	18.8%	25.7%	Not flagged

even balance between sensitivity (0.61) and specificity (0.64), but 0.26 is preferred because its higher sensitivity catches more flops, where approving flops is the more expensive mistake.

4.2.7 Hypothetical Film Pitch Test

The model estimates the chance that a film will flop based on its genre, cast size, budget, and whether it is a franchise or an original. For each pitch, it gives a risk estimate and a range that we can be 95% confident contains the true flop risk. Since 26% of films in our data flopped, any pitch with a higher estimated risk is flagged. For example, pitch B is flagged with an above-average 28% chance of flopping, and should therefore not be greenlit. Pitch A has a low 10% chance of flopping and should be greenlit. Pitch C's 22% chance is close to the average, so it should be greenlit with caution. Because our cut-off flags any film with above-average risk, Pitch C should still be reasonable to produce.

However, the model cannot reliably predict whether an individual film will flop. It only uses four factors, which explain a small part of why films flop. Other factors, such as script quality, casting star power and production quality are likely to matter more but are difficult to measure objectively. The model also predicts general patterns in flop trends, which means past trends are not indicative of future outcomes. Hence, it should be used as a supplementary factor alongside industry expertise to evaluate the film.

Appendix

Group number - 036

Members:

1. Aryaman Dhindsa - 35683422
2. Alexander Delamer - 36749346
3. Phi Vuong - 36127841

Link to Github Repo

This is the link to our github repo so that you can see we contributed equally to the project.

Task allocation

Task	Analysts	Reviewers	Target completion date	Completion type
Task 1	Alex	Aryaman	15th October	Draft
Task 2	Aryaman	Phi	16th October	Draft
Task 3	Phi	Aryaman, Alex	16th October	Draft
Task 4	Phi	Aryaman, Alex	20th October	Draft
Task 1	Alex	Aryaman, Phi	22th October	Final
Task 2	Aryaman	Phi, Alex	22th October	Final
Task 3	Phi	Aryaman, Alex	22th October	Final
Task 4	Phi, Aryaman, Alex	Phi, Aryaman, Alex	23th October	Final
Appendix	Aryaman	Phi, Alex	22th October	Draft
Final merging, formatting & reproducibility check	Alex	Phi, Aryaman	23th October	Final
Appendix	Aryaman	Phi, Alex	23th October	Final

Contribution Summary

This section be seen on github commit history and every member of the team confirms with this sentence that they have put approximately equal number of hours to create this project and are satisfied with the team effort.

Member name	Specific tasks performed	Hours contributed (approx.)
Alex	Task-1, Task-3,4 check, formatting and final checks	17 hrs
Aryaman	Task-2, Task-1,3,4 check, Appendix	16 hrs
Phi	Task-3 and 4, Task-2 check, final checks	18 hrs

Cross-review table

By signing below, you certify that you have reviewed the work of your teammates and that the final report represents a unified effort.

Task	Reviewed by	Key feedback provided
Task 1	Aryaman	Asked Alex to shorten the data description and cleaning process, additionally asked him to make some helpful plots that i can reference in task-2
Task 2	Phi	Change some of the formatting and task description, change the task flow and plots to match task-1
Task 3	Alex, Aryaman	Include more inline codes to cut some code chunks in order to not exceed 3 page limit, Add more comments in code chunks and format according to the task-1 and task-2

Task	Reviewed by	Key feedback provided
Task 4	Alex, Aryaman	Include more inline codes to cut some code chunks in order to not exceed 3 page limit, add some statement describing the differences between frequentist and baysian modeling